ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

6031/3

PAPER 3

JUNE 2020 SESSION 2 hours 30 minutes

Additional materials
Answer paper
Data Booklet
Electronic calculator

TIME: 2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **six** questions.

Answer **two** questions from Section A, **one** question from Section B, **two** questions from Section C and **one** question from section D.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 11 printed pages and 1 blank page.

Section A

Answer any ***two*** *questions from this section*

1 **(a)** An aldehyde, RCHO, reacts with acidified sodium cyanide according to the equation

$$RCHO + H^+ + CN^- \rightarrow RCH(OH)CN$$

Table 1.1 shows results of four experiments that were carried out to determine the kinetics of the reaction.

Table 1.1

experiment number	**[RCHO]/ moldm⁻³**	**[H⁺]/moldm⁻³**	**[CN⁻]/ moldm⁻³**	**relative initial rate/moldm⁻³s⁻¹**
1	0.10	0.30	0.30	5.00
2	0.10	0.25	0.25	4.16
3	0.10	0.25	0.30	5.00
4	0.125	0.25	0.25	5.20

(i) Deduce the order of reaction with respect to

1. RCHO

2. H^+

3. CN^-

(ii) Calculate the value of the rate constant, K.

[6]

(b) (i) Calculate the pH of 0.5 $moldm^{-3}$ benzoic acid.

[K_a of benzoic acid is 6.3×10^{-5} mol^2dm^{-6}]

(ii) Explain how a mixture of benzoic acid and sodium benzoate preserves food.

[6]

(c) Describe and explain the observations made when iron filings are added to a solution of tin (IV) ions.

[3]

Sn^{4+} (Fe^{2+}) Fe^{3+}

[Total: 15]

Fe^{3+} Sn^{2+} Fe^{2+}

Scanned by CamScanner

2 (a) Define the term

(i) *atomic radius*,

(ii) *shielding effect*. [2]

(b) Explain why

(i) the atomic radius of elements decrease across a period,

(ii) the cationic radius decrease from Na^+ to Al^{3+},

(iii) the anionic radius of an element is greater than the cationic radius,

(iv) the radius of the anion P^{3-} is greater than the radius of its non-metallic element P,

(v) there is an increase in atomic radius from Cl to Ar,

(vi) PCl_5 is a solid whereas PCl_3 is a liquid at r.t.p.

(vii) phosphonium ion, PH_4^+, is thermally unstable compared to the ammonium ion, NH_4^+. [7]

(c) $NOCl$ decomposes according to the equation.

$$2NOCl_{(g)} \rightleftharpoons 2NO_{(g)} + Cl_{2(g)}$$

A flask containing 0.50 moles of $NOCl$ was allowed to reach equilibrium at a pressure of 2.16 atmospheres. The equilibrium amount of $NOCl$ was found to be 0.30 moles.

Calculate the

(i) equilibrium amounts of

1. NO,

2. Cl_2,

(ii) equilibrium partial pressures of each of the three gases in the equilibrium mixture.

(iii) value of K_p.

[6]
[Total: 15]

3 **(a)** A mass of 2.00 g of an alloy of copper were dissolved in 50.00 cm^3 of concentrated nitric acid. The solution was made up to 200.00 cm^3 with distilled water. 20.00 cm^3 of this solution were reacted with aqueous potassium iodide. The iodine liberated reacted with 25.00 cm^3 of 1×10^{-1} $moldm^{-3}$ sodium thiosulphate.

Calculate the

(i) number moles of I_2 liberated by 20.00 cm^3 of $Cu^{2+}_{(aq)}$.

(ii) number moles of copper in 200.00 cm^3 of $Cu^{2+}_{(aq)}$.

(iii) percentage by mass of copper present in the alloy.

[7]

(b) **Table 3.1** shows values of enthalpies that describe the changes that occur when a solid $YX_{2(s)}$ dissolves in water

Table 3.1

enthalpy	**value**
lattice	$-2\,327$ KJ mol^{-1}
hydration of Y^{2+}	$-1\,920$ KJ mol^{-1}
hydration of X^-	-314 KJ mol^{-1}

(i) Define the term *enthalpy of hydration*

(ii) Calculate the enthalpy change of solution of $YX_{2(s)}$.

[3]

(c) Explain why plastic containers are used when preparing dilute sulphuric acid from the concentrated sulphuric acid in the laboratory.

[2]

(d) Carbonated drinks are preserved with $CO_{2(g)}$ dissolved under high pressure to produce the dynamic equilibrium:

$$CO_{2(g)} + H_2O_{(l)} \rightleftharpoons H_2CO_{3(aq)}$$

Explain why

(i) high pressure is necessary when sealing carbonated drinks,

(ii) rapid effervescence is observed when carbonated drinks are opened.

[3]
[Total: 15]

Section B

Answer any ***one*** *question from this section.*

4 **(a)** Explain why nitrogen

(i) is very unreactive under normal conditions,

(ii) reacts with oxygen in the air during thunderstorms.

[4]

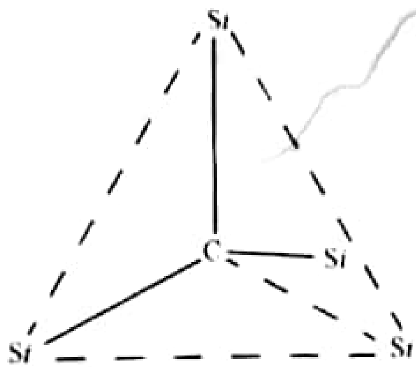
(b) **(i)** Describe how $NO_{2(g)}$ and $NO_{(g)}$ are produced during the combustion of fuel in a motor car engine.

(ii) Write chemical equations to show how $NO_{2(g)}$ acts as a catalyst in the formation of acid rain.

[7]

(c) Silicon carbide, SiC, has a structure shown in **Fig. 4** similar to that of diamond.

CO + NO2



C +

Fig. 4

Explain why SiC is used

(i) to make grinding and cutting tools,

(ii) as an insulator whereas graphite is an electrical conductor.

[4]
[Total: 15]

Scanned by CamScanner

MCl   AlCl₃ then Al(OH)₃ ...
Al₂O₃ +

(Al₂...)

X₂O₃ | these

5 (a) An element X, forms an oxide, X_2O_3 and a chloride, XCl_3. The oxide X_2O_3 is a solid at room temperature and is amphoteric.

(i) Write chemical equations to show how X_2O_3 reacts with

1. $H_2SO_{4(aq)}$,

2. $NaOH_{(aq)}$.

(ii) Explain how XCl_3 reacts with water. [4]

(b) (i) Describe and explain the trend in the thermal decomposition of Group two carbonates.

(ii) Give any two chemical equations that can be used to estimate the amount of calcium carbonate in a soil sample. [6]

(c) Construct balanced chemical equations for the

(i) oxidation of $Y_2O_3^{2-}$ ions by bromine to form YO_4^{2-} ions,

(ii) dissolution of the silver halide, AgX, in dilute ammonia,

(iii) reaction of sodium halide, NaX, with concentrated sulphuric acid to produce hydrogen sulphide gas.

[5]
[Total: 15]

these | X →

CaCO3

Section C

*Answer any **two** questions from this section.*

6 **(a)** The reaction scheme shown in **Fig. 6.1** shows how compound A can be converted to other organic compounds.

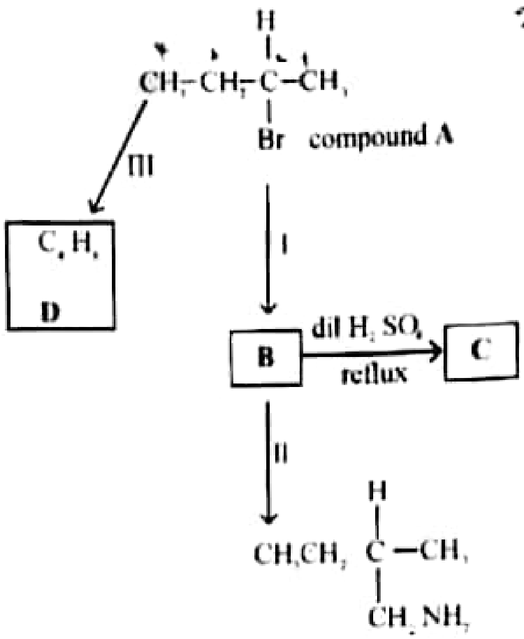


Fig. 6.1

(i) Name compound **A**.

(ii) Write reagents and conditions for reaction

1. I,

2. II.

(iii) Deduce the structural formula of

1. **B**,

2. **C**. [5]

(b) Reaction III forms, three isomeric alkenes of **D**.

(i) State reagents and conditions for reaction **III**.

(ii) Draw the structural formulae of the three isomers of **D**. [5]

(c) Another compound, **E**, which is isomeric with **D** is branched and does not exhibit *cis-trans* isomerism.

(i) Deduce the structural formula of compound **E**.

(ii) Draw the structural formula of the organic products produced when **E** reacts with

1. cold alkaline $KMnO_4$.

2. hot alkaline $KMnO_4$.

(iii) State the observations made in each of the reactions in **(ii)**. [6]
[Total: 15]

7 **(a)** (i) Draw the displayed formula for propylamine.

(ii) Comment on the acid-base nature of propylamine.

(iii) Suggest, with reasons, how the pH of propylamine compares with that of 2-aminopropanoic acid. [5]

(b) (i) Define the term *chiral molecule*.

(ii) State the type of isomerism shown by 2-aminopropanoic acid.

(iii) Draw the structures of the isomers. [4]

(c) (i) Explain the term *Zwitterion*.

(ii) Give the formula of the species of 2-amino propanoic acid in aqueous solution at

1. pH 2.

2. pH 12. [3]

(d) Describe and explain how 2-aminopropanoic acid can act as a buffer. [2]
[Total: 15]

8 **(a)** The structural formula of an organic pesticide is shown in **Fig. 8.1**.

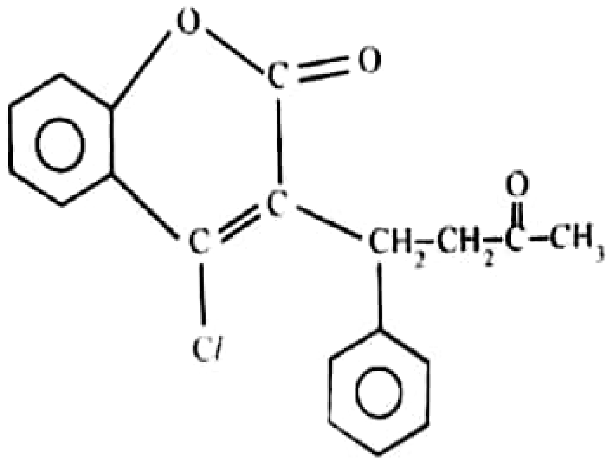


Fig. 8.1

(i) Deduce the molecular formula of the pesticide.

(ii) Identify any **three** functional groups present in the pesticide. [4]

(b) (i) Draw the structural formula of the organic products formed when the pesticide reacts with

1. Br_2 in an inert solvent,

2. PCl_5.

3. hot NaOH,

4. HCN.

(ii) Name the type of reaction undergone in each of the reactions in **(i)**. [8]

(c) A mass of *x* g of the pesticide reacted with sodium metal to produce 60.00 cm^3 of hydrogen gas.

Determine the value of *x*. [3]

[Total:15]

[Turn over

Section D

*Answer any **one** question from this section*

9 **(a)** **(i)** Distinguish between *bidentate ligand* and *polydentate ligand.*

(ii) Explain why $[Fe(H_2O)_6]^{2+}$ is paramagnetic and coloured whereas $[Fe(CN)_6]^{4-}$ is diamagnetic and colourless.

(iii) Draw the diamagnetic and paramagnetic ions of Fe^{2+}. [6]

(b) Describe and explain any one physical property of $[Cu(NH_3)_4(H_2O)_2]^{2+}$ and $[Cu(H_2O)_6]^{2+}$. [4]

(c) **(i)** Compare thin layer chromatography and gas liquid chromatography.

(ii) State any **two** applications of gas liquid chromatography. [5]

(d) Suggest why the use of nanomaterials in administering drugs is a better method in medicine and health compared to the present conventional methods. [2]

[Total: 15]

10 **(a)** A mixture of four amino acids, P, Q, R and, S was placed in the middle of a plate as shown in **Fig. 10.1**. The amino acids moved to positions shown in **Fig. 10.2** when electricity was switched on at pH 7.

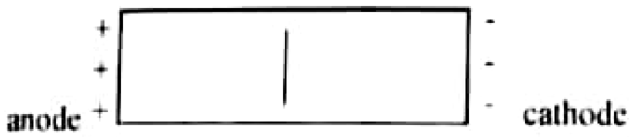
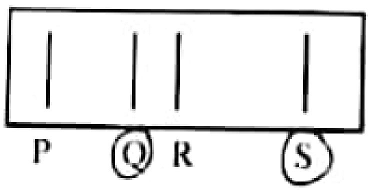


Fig. 10.1



NH_3^+

Fig. 10.2

(i) Name the technique used to separate the mixture.

(ii) State, with reasons, the

1. amino acid that predominantly exist as a Zwitterion.

2. difference between amino acid Q and S.

(iii) Explain why the results of the technique depends on pH.

(iv) Describe how the amino acids can be identified after the separation. [10]

(b) (i) Describe the process of removing the pollutant, sulphur dioxide, using the Fluid Gas Desulphurisation method.

(ii) Describe the effects of sulphur dioxide to the environment. [5]

[Total: 15]